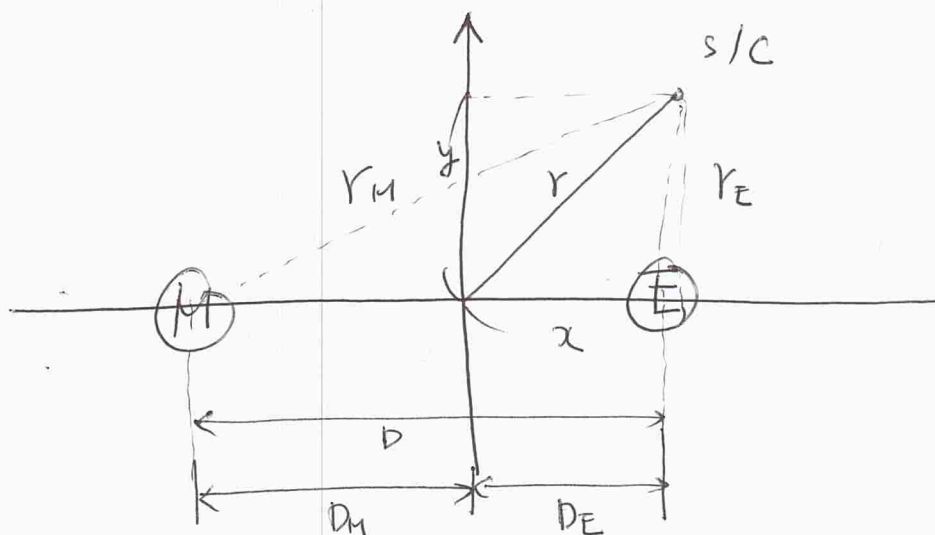


HW # 2

①



$$1. \begin{cases} \ddot{x} - 2\omega_0 \dot{y} - \omega_0^2 x = -\frac{\mu_E (x - D_E)}{|r_E|^3} - \frac{\mu_M (x + D_M)}{|r_M|^3} + \frac{F_x}{m} \\ \ddot{y} + 2\omega_0 \dot{x} - \omega_0^2 y = -\frac{\mu_E y}{|r_E|^3} - \frac{\mu_M y}{|r_M|^3} + \frac{F_y}{m} \\ \ddot{z} = -\frac{\mu_E z}{|r_E|^3} - \frac{\mu_M z}{|r_M|^3} + \frac{F_z}{m} \end{cases}$$

Where

$$\mu_E = 3.9860 \times 10^5 \text{ km}^3/\text{s}^2$$

$$\mu_M = 4.887 \times 10^3 \text{ km}^3/\text{s}^2$$

$$D_E = 4.674 \times 10^3 \text{ km}$$

$$D_M = 3.8007 \times 10^5 \text{ km}$$

$$D = D_E + D_M = 3.8474 \times 10^5 \text{ km}$$

$$\omega_0 = \sqrt{\frac{G(M_M + M_E)}{D^3}} = \sqrt{\frac{\mu_M + \mu_E}{D^3}} = 2.6620 \times 10^{-6} \text{ rad/sec}$$

$$r_E^2 = (x - D_E)^2 + y^2 + z^2$$

$$r_M^2 = (x + D_M)^2 + y^2 + z^2$$

(2)

$$2. \quad \underline{(x, y, z)_{L_2} = (-4.4465 \times 10^5 \text{ km}, 0, 0)}$$

Since $(x, y, z)_{L_2}$ is an eq. point,

$$(\dot{x}, \dot{y}, \dot{z})_{L_2} = (\ddot{x}, \ddot{y}, \ddot{z})_{L_2} \equiv 0$$

$$\textcircled{1} \quad -\omega_0^2 x = -\frac{\mu_E (x - D_E)}{(\sqrt{(x - D_E)^2 + y^2 + z^2})^3} - \frac{\mu_M (x + D_M)}{(\sqrt{(x + D_M)^2 + y^2 + z^2})^3}$$

$$3.1502 \times 10^{-6} \approx 3.1461 \times 10^{-6} \text{ (km/s}^2\text{)}$$

② & ③ are also satisfied with the information above.

$\therefore (x, y, z)_{L_2}$ is in equilibrium at L_2

If the origin of the inertial frame is located at the barycenter, ~~for~~ then L_2 is circling around the barycenter with angular frequency

$$\underline{\omega_0 = 2.6620 \times 10^{-6} \text{ rad/sec and the period}}$$

$$\underline{T = 27.3218 \text{ days}}$$

Mathematically

$$r_{L_2} = x_{L_2} i$$

$$\dot{r}_{L_2} = \dot{x}_{L_2} i + \omega_0 k \times x_{L_2} i$$

$$= x_{L_2} \omega_0 j \quad \curvearrowright$$

3. Linearization about L_2

$$\begin{aligned} \Delta \ddot{x} - 2\omega_0 \Delta \dot{y} - 7.3809\omega_0^2 \Delta x &= \frac{F_x}{m} \\ \Delta \ddot{y} + 2\omega_0 \Delta \dot{x} + 2.1904\omega_0^2 \Delta y &= \frac{F_y}{m} \\ \Delta \ddot{z} + 3.1904\omega_0^2 \Delta z &= \frac{F_z}{m} \end{aligned} \quad \left. \begin{array}{l} \text{In-Plane} \\ \text{Out-of-Plane} \end{array} \right\}$$

Let the state variable

$$\begin{aligned} & [\Delta x \quad \Delta \dot{x} \quad \Delta y \quad \Delta \dot{y} \quad \Delta z \quad \Delta \dot{z}] \\ &= [\Delta x \quad \Delta \dot{x} \quad \Delta y \quad \Delta \dot{y} \quad \Delta z \quad \Delta \dot{z}] \end{aligned}$$

Then

$$\begin{bmatrix} \Delta \dot{x} \\ \Delta \ddot{x} \\ \Delta \dot{y} \\ \Delta \ddot{y} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 7.3809\omega_0^2 & 0 & 0 & 2\omega_0 \\ 0 & 0 & 0 & 1 \\ 0 & -2\omega_0 & -2.1904\omega_0^2 & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta \dot{x} \\ \Delta y \\ \Delta \dot{y} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{F_x}{m} \\ \frac{F_y}{m} \end{bmatrix}$$

$$\begin{bmatrix} \Delta \dot{z} \\ \Delta \ddot{z} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -3.1904\omega_0^2 & 0 \end{bmatrix} \begin{bmatrix} \Delta z \\ \Delta \dot{z} \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \frac{F_z}{m}$$

In-Plane Eigenvalue: $\pm 2.1587\omega_0$
 $\pm j 1.8627\omega_0$

Out-of-Plane Eigenvalue $\pm j 1.7862\omega_0$

$$4. \begin{bmatrix} \Delta x \\ \Delta V_x \\ \Delta y \\ \Delta V_y \\ \Delta z \\ \Delta V_z \end{bmatrix}_{t=0} = \begin{bmatrix} 0 \\ -0.005966 \\ -3.55 \text{ km} \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Since $\begin{bmatrix} \Delta z \\ \Delta V_z \end{bmatrix}_{t=0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, the out-of-plane motion is in

equilibrium, i.e., there is no perturbed out-of-plane motion.

In-plane position/velocity vector histories are shown at page 5.

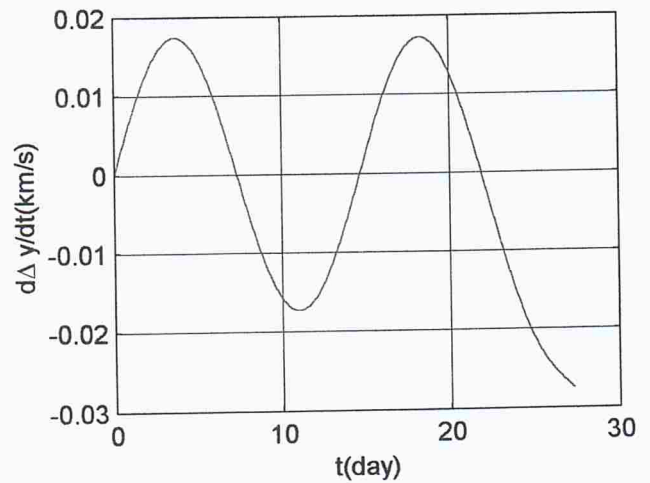
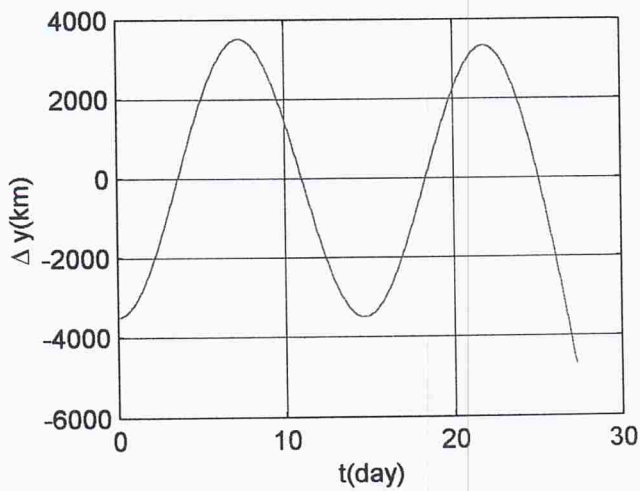
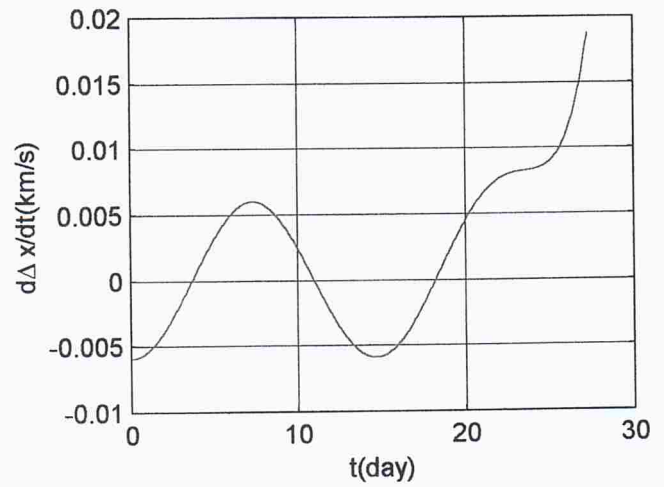
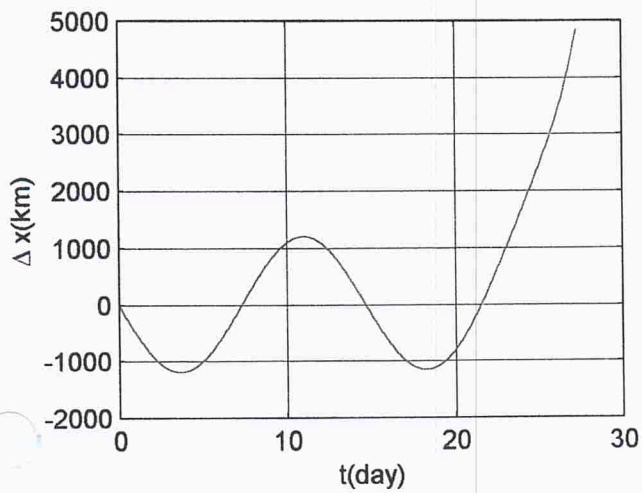
Δy vs. Δx during one orbital revolution of $T=27.3218$ days

is shown at Page 6. It is clear that the motion is NOT Halo Orbit.

The initial condition does instigate the unstable mode of uncontrolled dynamics, and the trajectory starts to diverge in about 20 days.

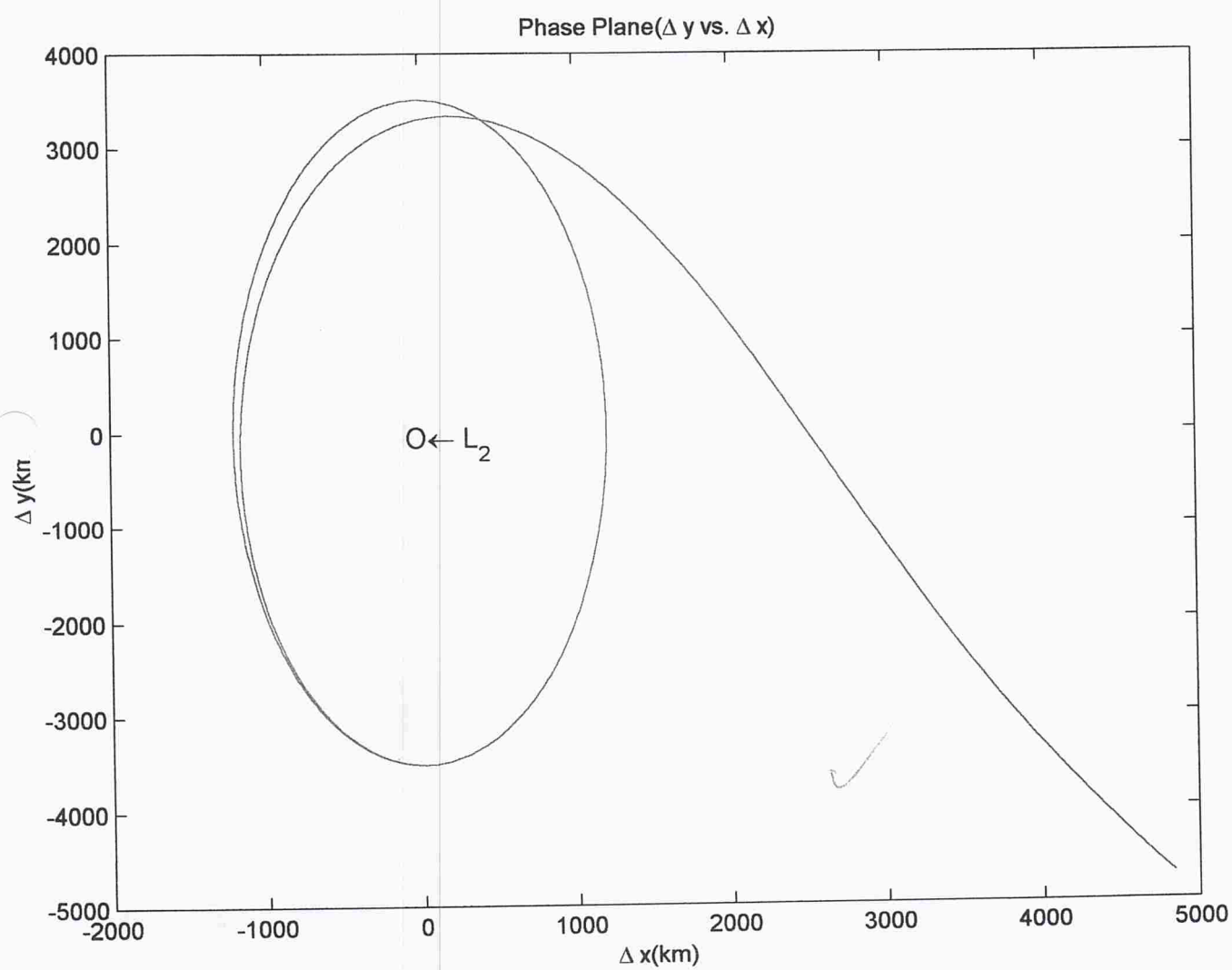
5

State History in 1 period $\approx 2^n$ days

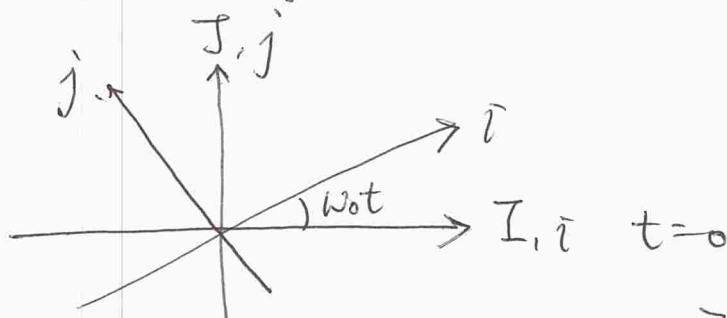


6

$t = [0, 27 \text{ days}]$



Finally to analyze the motion in the Inertial frame, let the Inertial frame XYZ be aligned with barycentric frame xyz



$$\begin{bmatrix} i \\ j \end{bmatrix} = \begin{bmatrix} \cos \omega_0 t & \sin \omega_0 t \\ -\sin \omega_0 t & \cos \omega_0 t \end{bmatrix} \begin{bmatrix} I \\ J \end{bmatrix}$$

Then, the motion of S/C in the Inertial frame is given by

$$\vec{r} = \vec{r}_{L_2} + (\Delta x(t) \hat{i} + \Delta y(t) \hat{j})$$

$$= x_{L_2} \hat{i} + \Delta x(t) \hat{i} + \Delta y(t) \hat{j}$$

$$= (x_{L_2} + \Delta x(t)) \left(\frac{\cos \omega_0 t}{\cos} I + \sin \omega_0 t J \right)$$

$$+ \Delta y(t) (-\sin \omega_0 t I + \cos \omega_0 t J)$$

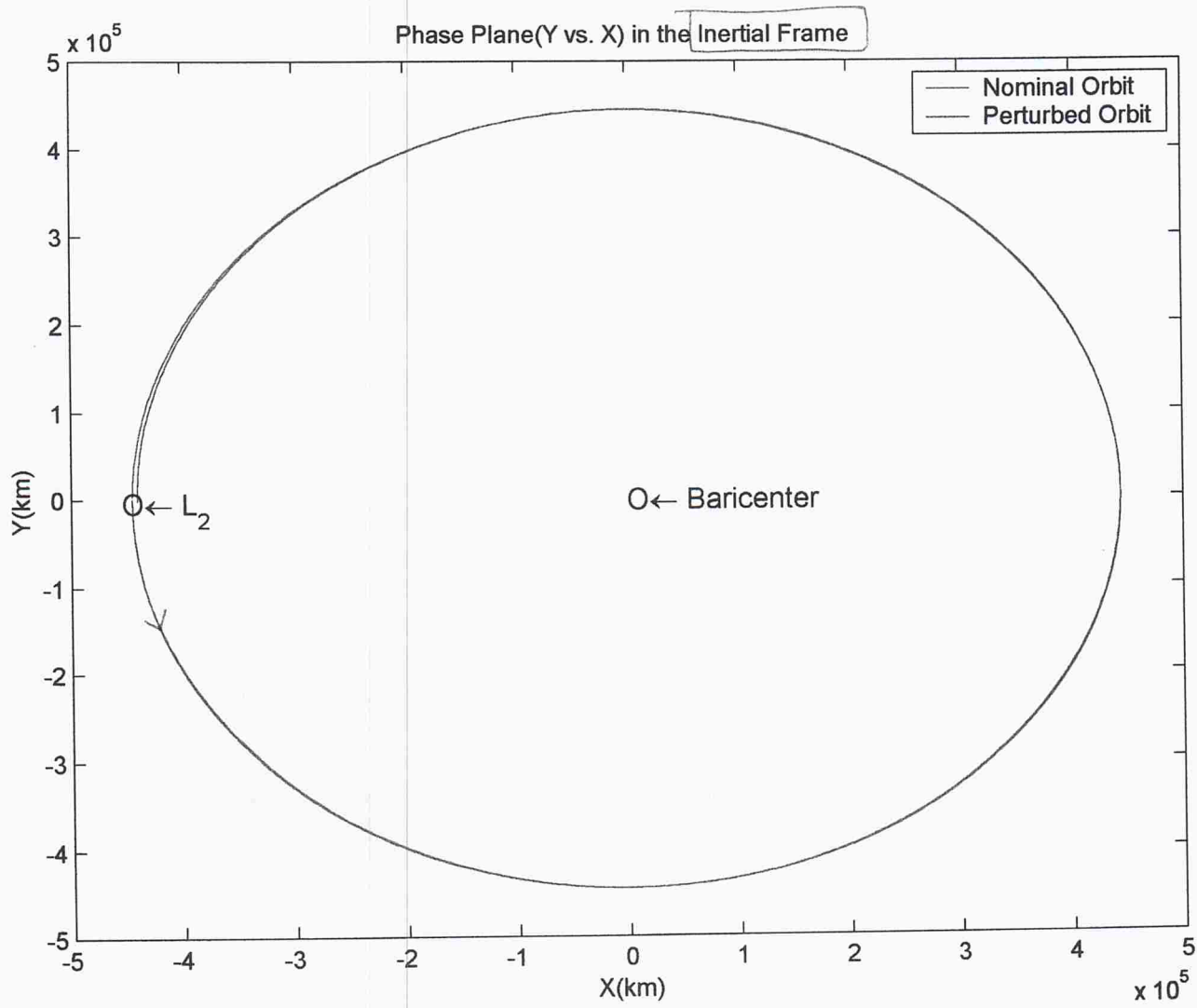
$$= \left[(x_{L_2} + \Delta x(t)) \cos \omega_0 t - \Delta y(t) \sin \omega_0 t \right] I$$

$$+ \left[(x_{L_2} + \Delta x(t)) \sin \omega_0 t + \Delta y(t) \cos \omega_0 t \right] J$$

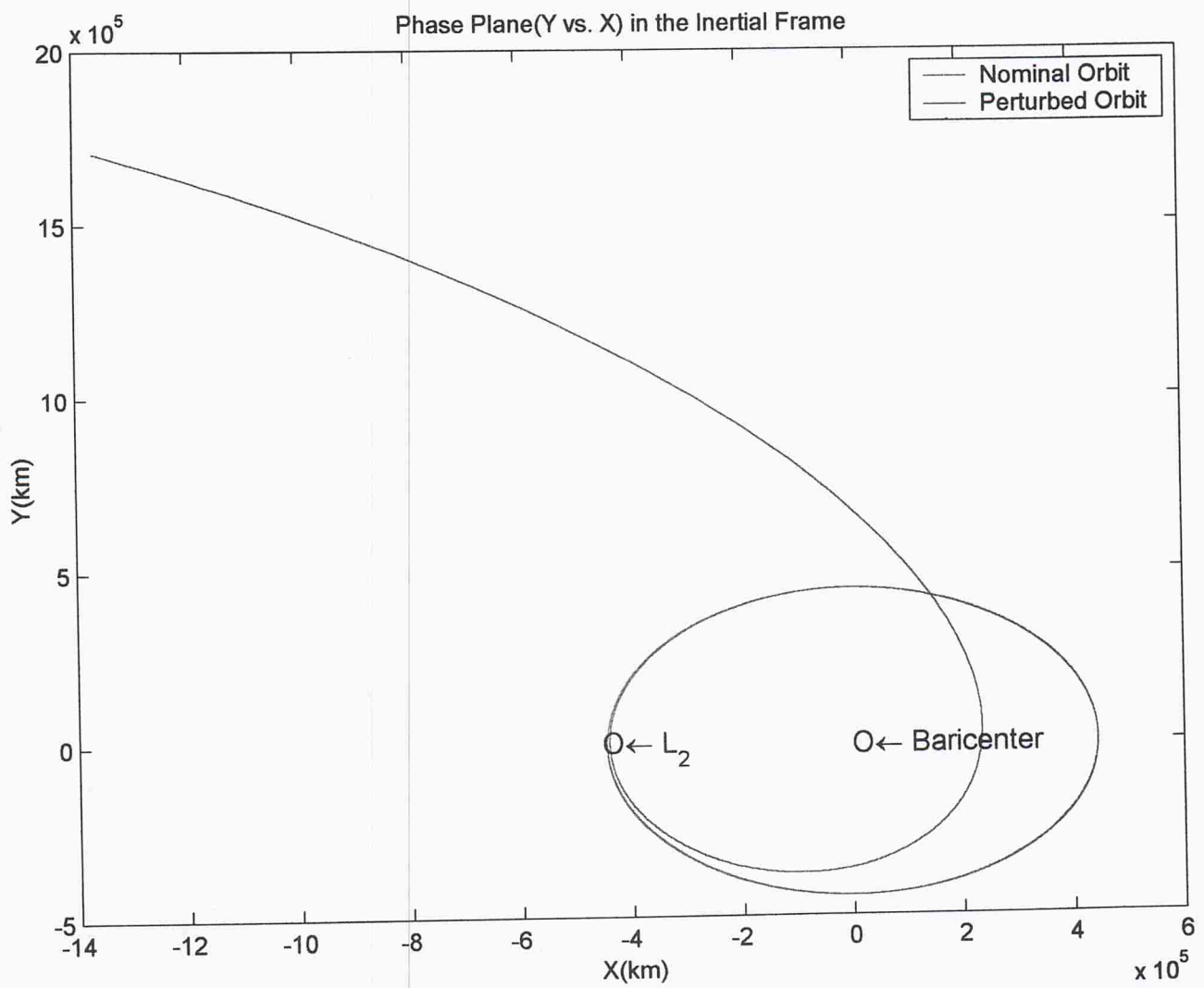
$$\begin{aligned} \dot{\vec{r}} = & \left[\Delta \dot{v}_x(t) \cos \omega_0 t - (x_{L_2} + \Delta x(t)) \omega_0 \sin \omega_0 t \right. \\ & \left. - \Delta \dot{v}_y(t) \sin \omega_0 t - \Delta y(t) \omega_0 \cos \omega_0 t \right] I \\ & + \left[\Delta \dot{v}_x(t) \sin \omega_0 t + (x_{L_2} + \Delta x(t)) \omega_0 \cos \omega_0 t \right. \\ & \left. + \Delta \dot{v}_y(t) \cos \omega_0 t - \Delta y(t) \omega_0 \sin \omega_0 t \right] J \end{aligned}$$

8

$t \in [0, 27 \text{ days}]$



✓

$t \in [0, 40 \text{ days}]$ 

5. For the In-Plane dynamics

$$\text{rank}([B \ AB \ A^2B \ A^3B]) = 4$$

$\Rightarrow$ completely controllable with two in-plane force components

For the out-of-plane dynamics

$$\text{rank}([B \ AB]) = \text{rank} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = 2$$

$\Rightarrow$ Completely Controllable. with one out-of-plane force component.

6. (a) ~~Since $\Delta \vec{z} = \Delta \dot{\vec{z}} =$~~

Since $\Delta z(t_f) = \Delta \dot{z}(t_f) = 0$, out-of-plane motion is stationary, and we only consider in-plane motion

$$\begin{array}{c}
 \begin{bmatrix} \Delta z \\ \Delta v_y \\ \Delta y \\ \Delta v_x \end{bmatrix} \xrightarrow[t=0]{L_2} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ v_x^+ \\ 0 \\ v_y^+ \end{bmatrix} \xrightarrow[t=t_0^+]{STM} \begin{bmatrix} \Delta x(t_f)=0 \\ \cancel{\Delta y(t_f)}? \\ \Delta y(t_f)=-3505 \\ ? \end{bmatrix} \xrightarrow[t=t_f=2.5 \text{ days}]{\begin{pmatrix} v_x^- \\ v_y^- \end{pmatrix}} \begin{bmatrix} 0 \\ -0.005966 \\ -3.505 \\ 0 \end{bmatrix}
 \end{array}$$

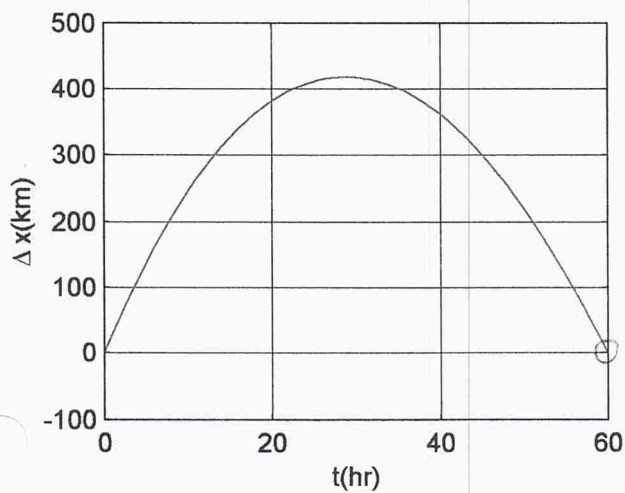
$$\Delta V_0 = 0.0084 \hat{i} - 0.0169 \hat{j} \text{ km/s}$$

$$\Delta V_f = 0.0012 \hat{i} + 0.0106 \hat{j} \text{ km/s}$$

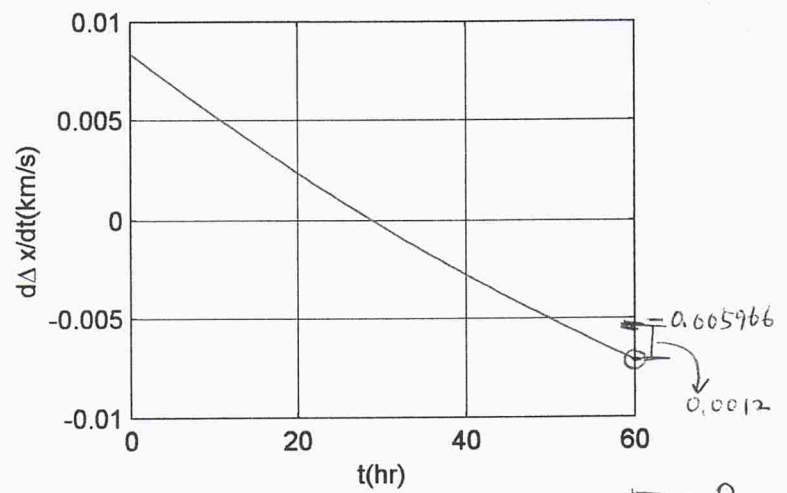
12

2

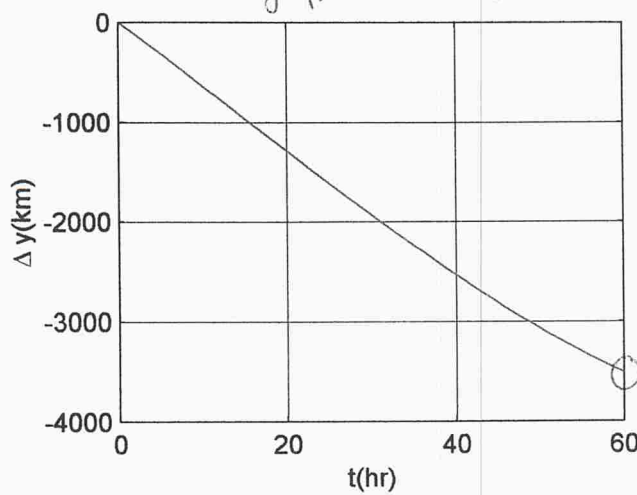
$$\Delta x(t_f) = 0$$



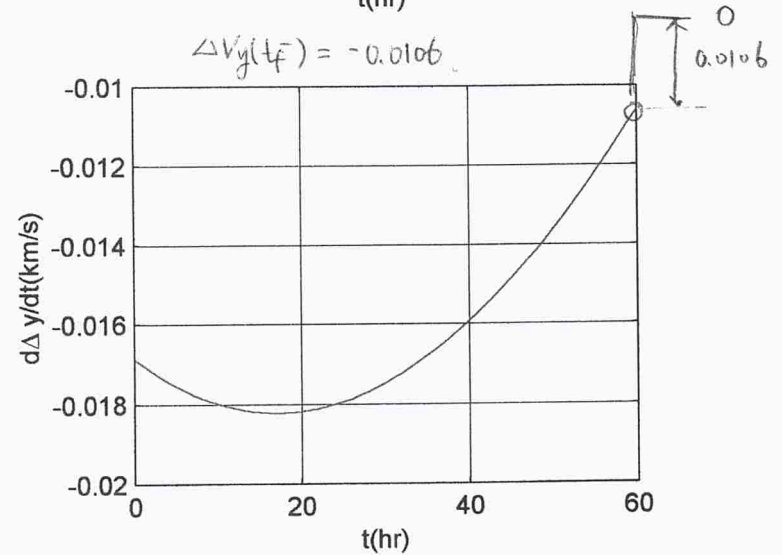
$$\Delta V_x(t_f^-) = -0.0092$$



$$\Delta y(t_f) = -3505$$



$$\Delta V_y(t_f^-) = -0.0106$$



(b) Among many options, the optimal maneuver is chosen which was treated in class. That is

$$\begin{bmatrix} F_x(t)/m \\ F_y(t)/m \end{bmatrix} = B^T e^{A^T(t-t_f)} W(0, t_f)^{-1} \begin{bmatrix} \Delta x(t_f) \\ \Delta v_x(t_f) \\ \Delta y(t_f) \\ \Delta v_y(t_f) \end{bmatrix} - e^{At_f} \begin{bmatrix} \Delta x(0) \\ \Delta v_x(0) \\ \Delta y(0) \\ \Delta v_y(0) \end{bmatrix}$$

The following plots shows that this control scheme achieves the desired maneuver.

(b') Using piecewise constant maneuver.

$$\frac{\vec{F}}{m} = \begin{cases} (1.058 \hat{i} - 2.869 \hat{j}) \times 10^{-7} \text{ N/kg} & 0 \leq t \leq 1.08 \times 10^5 \text{ sec} \\ (-0.0072 \hat{i} + 2.33 \hat{j}) \times 10^{-7} \text{ N/kg} & 1.08 \times 10^5 \text{ sec} < t < 2.16 \times 10^5 \text{ sec} \end{cases}$$

